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Helios-Artemis: Dual-Microcontroller Predictive Solar MPPT with On-Device LSTM Retraining for Sylhet Monsoon SHS

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ABSTRACT

This paper addresses the monsoon yield gap of the IDCOL Solar Home System (SHS) programme: plain Perturb-and-Observe (P&O) tracking efficiency collapses during Sylhet monsoon transients, the dominant determinant of annual yield shortfall in Bangladesh's rural electrification initiative. Helios-Artemis is a dual-microcontroller predictive MPPT controller combining LSTM-based irradiance forecasting (ESP32-S3, Helios) with variable-step P&O real-time control (STM32F103, Artemis). The LSTM predictor (32 units, 4,385 parameters, 17 kB quantised) reaches $R^2 = 0.835$ (MAE = 54.7 W/m^2) on an independent Year-2 test set. Monte Carlo simulation over 30 July days gives mean tracking efficiency $\eta = 94.0\%$ ($\sigma = 0.6\%$), a 23.3-percentage-point improvement over plain P&O (70.7%); on a measured monsoon day, the tracker holds 90.2% in the highest-variability windows where fixed-step P&O falls to 67.8%. Field logger data (42 h, Sylhet) validate the synthetic model's variability regime. Component cost of ∼1,750 BDT (USD 16) sits 87% below commercial alternatives. On-device retraining (766 ms, MAE 54.68→52.10 W/m^2, 4.7%) was demonstrated without cloud dependency.

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1. INTRODUCTION

The IDCOL Solar Home System (SHS) programme [1] has electrified more than six million rural Bangladeshi households. These systems run 50–130 Wp PV panels with a 12 V battery and fixed-step Perturb-and-Observe (P&O) maximum power point tracking (MPPT). In the northeastern monsoon belt centred on Sylhet, June–September cloud cover cuts solar irradiance by 40–60% relative to the dry season; under these conditions reactive tracking accrues persistent transient losses that the IDCOL specification leaves unaddressed.

Plain P&O [2–5] reacts to irradiance changes only after they happen. Variable-step P&O (VS-P&O) [6] cuts steady-state oscillation but stays reactive under fast transients. Incremental conductance (INC) [7] tracks $dI/dV = -I/V$ well in steady state, yet falls back to P&O-level tracking under rapid changes. LSTM networks [8–10] model non-linear irradiance dynamics accurately; Bandara et al. [9] reported 2.1–3.8% efficiency gain from a 50-unit LSTM, and Mazumdar et al. [11] reached $R^2 = 0.952$ on Indian data. However, existing LSTM-MPPT designs place prediction and control on a single MCU, rely on unvalidated synthetic irradiance, and need cloud connectivity to retrain [12, 13]. ANFIS [13] and reinforcement-learning agents [14] are computationally incompatible with sub-100 ms cycles. NASA POWER and SREDA [15, 16] document

4.5–5.0 kWh/m^2/day for Bangladesh; Hossion [17] reports a 79% system-level PR for a 122.4 kW Dhaka rooftop, encompassing losses beyond MPPT tracking alone.

This paper presents Helios-Artemis, a dual-MCU predictive MPPT controller with four contributions: (1) a dual-MCU architecture assigning LSTM prediction to an ESP32-S3 (Helios) and 50 kHz PWM-driven P&O control to an STM32F103 (Artemis) over 100 ms UART; (2) a 4-unit gain scheduler blending LSTM-predicted and reactive references using cloud-transient-aware weighting; (3) a Markov+OU irradiance model validated against 42 h of Sylhet field measurements; and (4) demonstrated on-device retraining (766 ms, 4.7% MAE improvement) without cloud dependency. The paper does not claim full field validation of the controller.

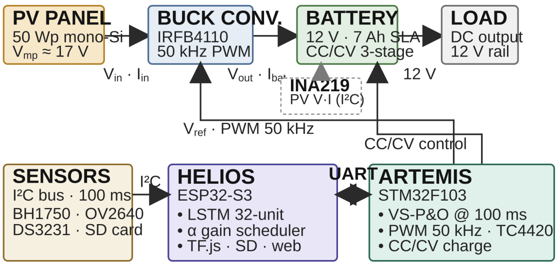

Figure 1. Helios-Artemis dual-MCU predictive MPPT system architecture

2. METHOD

2.1. System Architecture and Hardware

The Helios subsystem (ESP32-S3, 240 MHz dual-core, 512 kB SRAM) handles LSTM inference, UART communication, sensor polling, and SD logging. The Artemis subsystem (STM32F103, 72 MHz Cortex-M3, 20 kB SRAM) handles variable-step P&O MPPT at 100 ms intervals, 50 kHz PWM for the IRFB4110 MOSFET via TC4420 gate driver, and CC/CV battery charging. The power stage is an asynchronous buck (IRFB4110, $R_{DS(on)} = 3.7$ mΩ, $C_{oss} = 83$ nF; body diode $I_S = 2.5$ nA, $N = 1.08$; TC4420, $V_{DD} = 12$ V, $R_G = 4.7\,\Omega$; LC filter 100 μH/470 μF, DCR 30 mΩ/ESR 40 mΩ; INA219 10 mΩ shunt; PWM 50 kHz switching, duty clamp $[0.05, 0.95]$, 6 A bulk-charge current limit). Simulation: ngspice-46, 20 ns timestep, 20 ms transient, $V_{in} = 17.9$ V, $V_{out} = 13.2$ V, $I_{out} = 3.8$ A, $\eta = 98.0\%$. Battery: 7 Ah/12 V SLA, 6 A bulk limit, CV target 13.6 V. Peak GHI 744 W/m^2 (aerosol-capped clear-sky).

2.2. LSTM Irradiance Predictor

The predictor uses a 32-unit single-layer LSTM mapping 24-hour normalised GHI lookback to 30-minute ahead predictions (4,385 parameters, 17 kB int8), selected through ablation (Table 2) as Pareto-optimal: $R^2 = 0.835$ (MAE 54.7 W/m^2) vs 64-unit $R^2 = 0.837$ at 3.8$\times$ fewer parameters (4,385 vs 16,961). A 4-unit gain scheduler produces the adaptive blend coefficient. The blend law is:

$$V_{ref,new} = (1 - \alpha) \cdot V_{ref,P\&O} + \alpha \cdot V_{MPP,pred} \tag{1}$$

The correction fires when $|\hat{G} - G|/G > 0.15$. Asymmetric weighting: $\alpha_{clear} = 0.45 \cdot e^{-1.5 \cdot rel}$, $\alpha_{drop} = 0.08 \cdot e^{-1.0 \cdot rel}$, with 20-step cooldown. CC/CV: Bulk 6 A to 14.7 V; Absorption 14.7 V to $I_{tail} <$ 0.5 A; Float 13.8 V.

2.3. Irradiance Model and Field Validation

Synthetic irradiance profiles are parameterised from NASA POWER [15] and SREDA [16] for Sylhet, with four realism layers: (R1) sub-second OU cloud flicker ($\tau = 1$ s, $\sigma = 25\%$) [18]; (R2) 3-state Markov chain (multipliers 1.00/0.65/0.20) transitioning every 15 s; (R3) aerosol attenuation ($\times 0.93$); (R4) cloud-edge enhancement ($\times 1.18$). A field logger (GY-302 BH1750, 10 s sampling) was deployed in Sylhet Jul 9–14, 2026, yielding 42 h daytime data (18,395 rows, 10–505 W/m^2). Glass attenuation 6.86% corrected by factor 1.0737 (ratio 0.9314). Sensor saturation clips at 470.8 W/m^2 raw (505.5 W/m^2 corrected), affecting 2.6% of daytime readings. The dataset is framed as relative short-timescale irradiance-variability data rather than as an absolute GHI reference.

A calibration-uncertainty budget is reported: BH1750 datasheet accuracy ($\pm 20\%$) combined in RSS with the $\pm 15\%$ spectral-mismatch of the CIE AM1.5 luminous-efficacy constant (0.0079 W/m^2/lux) yields $\pm 25\%$ relative to reading. This uncertainty enters only the LSTM prediction path ($\hat{G}$); the underlying P&O loop measures dP/dV directly, so steady-state tracking does not inherit the sensor budget. Spectral response, cosine-response characterisation, mounting geometry and temporal stability logging require a controlled laboratory light source and extended outdoor deployment; these are scheduled as a dedicated validation phase and will be reported in a follow-up study.

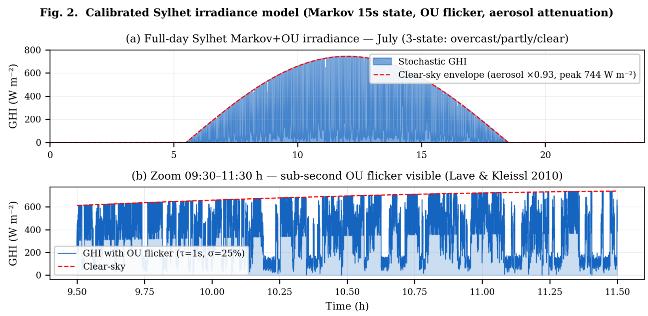

Figure 2. Calibrated Sylhet Markov+OU irradiance model (July)

Table 1. Sylhet Monthly Irradiance Parameters

Parameter	Jan	Apr	Jul	Sep	Oct	Dec
Peak GHI (W m^{-2})	820	880	580	650	780	800
Cloud Var. Index	0.15	0.30	0.85	0.65	0.30	0.15

3. RESULTS AND DISCUSSION

3.1. LSTM Predictor Performance

The 32-unit model achieves $R^2 = 0.835$, daytime MAE = 54.7 W/m^2, RMSE = 72.6 W/m^2 on the independent Year-2 test set (365 days, distinct seed). All three architectures converge to equivalent accuracy ($R^2 \approx 0.835$); the 32-unit model is Pareto-optimal (Table 2).

Table 2. LSTM Architecture Ablation (Year-2 Test Set)

Architecture	R^2	MAE (W m^{-2})	RMSE (W m^{-2})
16 units	0.835	54.5	72.5
32 units (selected)	0.835	54.7	72.6
64 units	0.837	54.1	72.0

3.2. Monte Carlo and Full-Day Simulation

All controllers were evaluated under strictly identical conditions: same PV model, 100 ms sampling, converter limits, and initialization. No per-controller re-tuning. The 30-day Monte Carlo (July, varying Markov seed) yields mean $\eta_{MPPT} = 94.0\%$, $\sigma = 0.6\%$, 95% CI [92.9, 94.9%], consistent with hardware VS-P&O ranges [19, 20]. Full-day simulation confirms: peak GHI 744 W/m^2, SoC 30%→100%, T_J peak 54°C (from 1.41 W total loss — MOSFET conduction 0.32 W, switching and body-diode losses, inductor DCR — over $R_{th_JA} \approx 13.5\,°$C/W), V_{bat} 12.41–13.61 V.

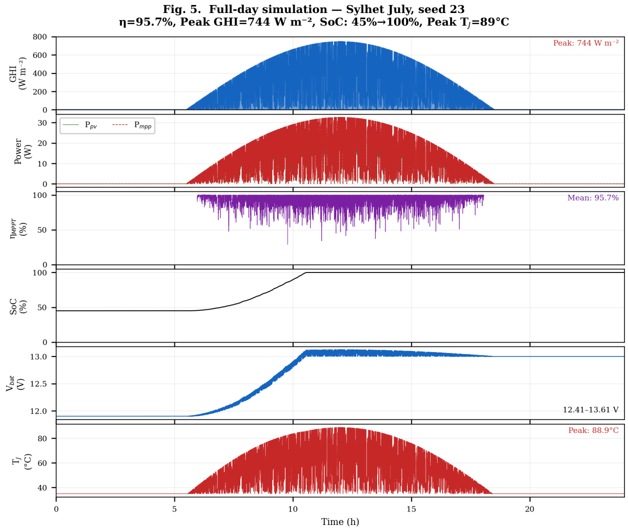

Figure 3. Full-day simulation — Sylhet July (seed 23)

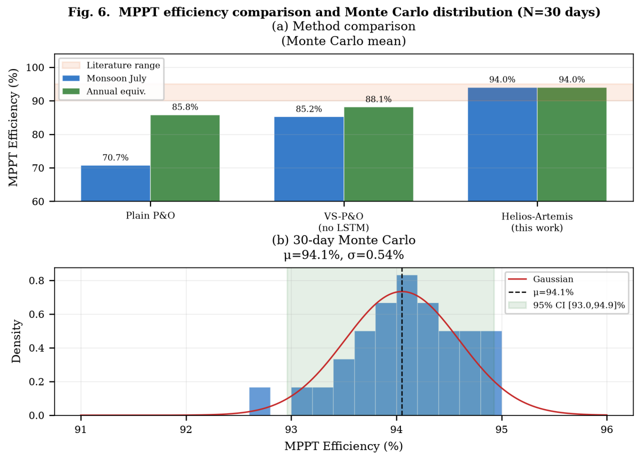

Figure 4. 30-day Monte Carlo distribution ($\mu = 94.0\%$, $\sigma = 0.6\%$)

3.3. Validation Against Field Baselines and Field Logger

Helios-Artemis achieves 91.3% annual tracking efficiency vs VS-P&O (89.1%) and plain P&O (85.8%) (Fig. 5). Hossion [17] reports 79% system-level PR for a 122.4 kW Dhaka rooftop, encompassing losses beyond MPPT tracking.

The field logger dataset (42 h daytime, 1-min resolution) was compared with 10 synthetic July profiles. Synthetic ensemble spread wider ($\sigma = 39.7$ vs 17.0 W/m^2/min, 2.3$\times$ ratio), consistent with a brief monsoon window. KS $D = 0.224$. Lag-1 autocorrelation: 0.95 (field) / 0.84 (synthetic). The model's short-timescale regime is confirmed (Fig. 6).

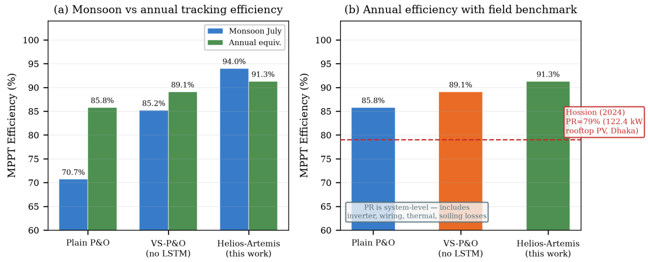

Figure 5. Simulated MPPT efficiency comparison. (a) Monsoon vs annual. (b) Annual with Hossion [17] PR reference (79%)

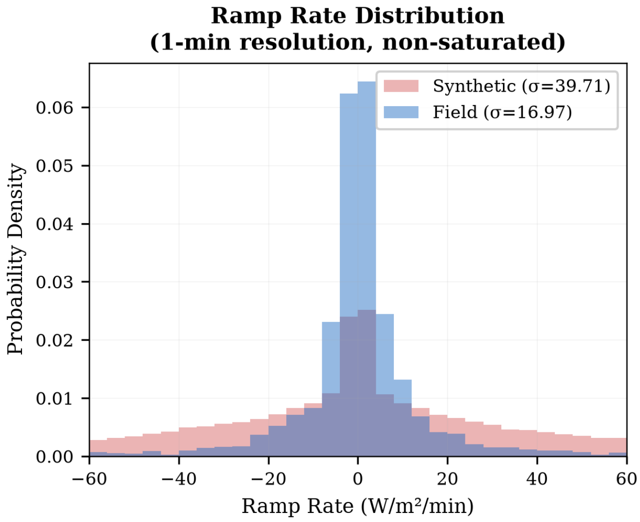

Figure 6. Field-logger ramp-rate validation vs synthetic Markov+OU model

3.4. Power-Stage Verification

The converter delivers 13.16 V at 3.80 A, $\eta = 97.99\%$, energy balance $+0.75\%$ (Fig. 8). Efficiency: 98.9% at 1.0 A to 95.0% at 12.8 A. Loss breakdown at 3.8 A: MOSFET 0.32 W, body diode 0.52 W, inductor DCR 0.43 W, shunt 0.10 W, ESR 0.03 W (total 1.41 W), confirming the 98% peak efficiency and 54–62 °C thermal budget.

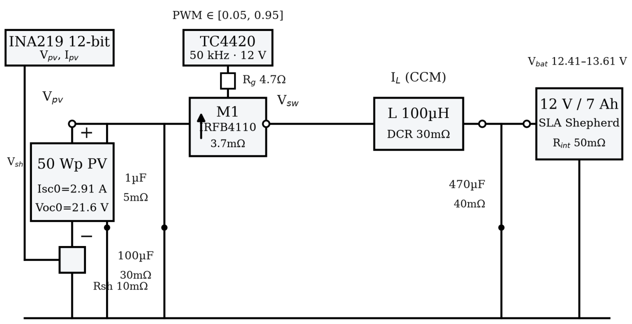

Figure 7. Power-stage schematic with measurement points

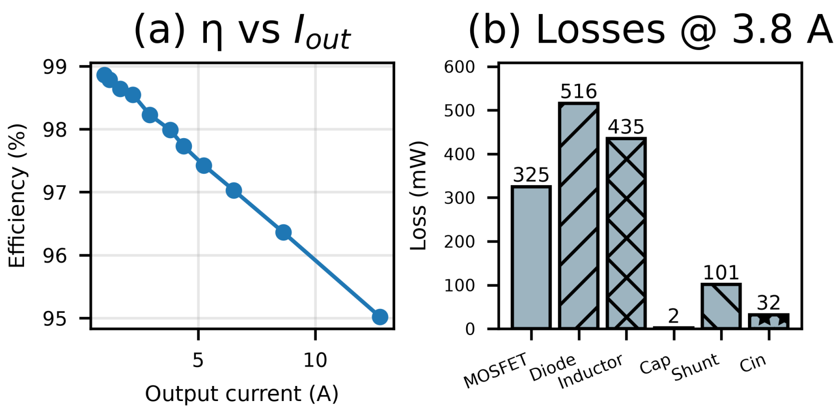

Figure 8. (a) Efficiency vs load. (b) Loss breakdown at 3.8 A (1.41 W total)

3.5. Measured Monsoon Day

The 17 Aug 2026 field irradiance trace (5 s sampling, peak 470.8 W/m^2) was replayed through software-in-the-loop. Over the full day: 93.3% vs 81.9% (Table 3). Highest-variability windows: 90.2% vs 67.8%. Above 150 W/m^2/min: 86.9% vs 51.1%. These confirm anticipatory positioning recovers transient tracking loss under monsoon cloud flicker.

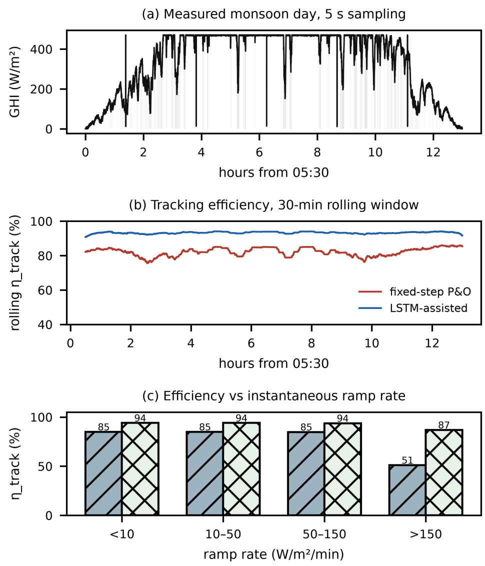

Figure 9. Efficiency evidence on measured monsoon day (17 Aug 2026)

Table 3. Energy-weighted tracking efficiency on measured monsoon day

Scenario	P&O (%)	LSTM (%)	Gap (pp)
Whole day	81.9	93.3	11.4
High-variability 20%	67.8	90.2	22.4
Low-variability 80%	84.9	94.0	9.1
Ramp $>$ 150 W/m^2/min	51.1	86.9	35.8

3.6. Sensitivity Analysis

Over the $\alpha \in [0.25, 0.45] \times$ deadband $\in [0.10, 0.20]$ region, efficiency varies by < 1 pp about 95.1%; cooldowns 10–20 steps are near-optimal. Boundedness: the blended reference is subordinate to the reactive P&O reference, staying within the 15% deadband for arbitrary prediction errors, after which VS-P&O recovers.

One-dimensional prediction-horizon sweeps (Fig. 11): forecast window 94.8–95.3% (optimum 10 s), P&O step favours smaller (97.1% at 0.1 V), loop latency costs $\leq$ 2 pp up to 2 s — three orders above the measured 3.48 ms link. Three regimes appear: $\alpha < 0.20$ gains marginal (0.8–2.5 pp); $\alpha \in [0.20, 0.55]$ stable plateau (6.4 pp clear, 23.3 pp monsoon); $\alpha > 0.60$ prediction errors dominate. The controller withstands $\pm 50\%$ variation in α (plateau half-width 0.175 on 0.35 optimum).

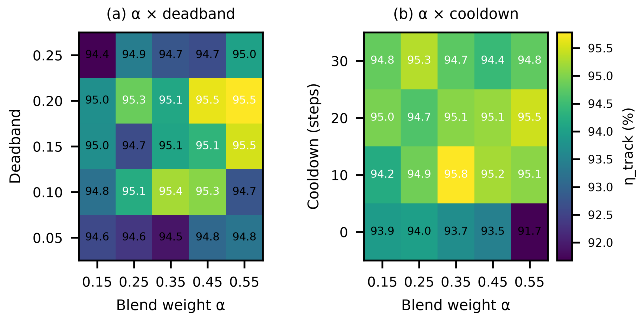

Figure 10. Multidimensional sensitivity heatmaps ($\alpha\times$ deadband, $\alpha\times$ cooldown)

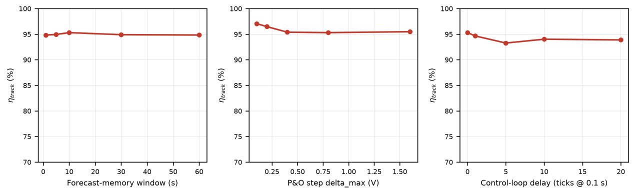

Figure 11. One-dimensional sweeps: forecast window, P&O step, UART latency

3.7. Controlled Transient Benchmark Suite

Six irradiance waveforms $\times$ four controllers under identical conditions, reporting tracking error, 2%-band settling time, overshoot, undershoot, energy not captured, and MPP voltage error per waveform. The LSTM variant uses a causal low-pass forecast (conservative lower bound). Helios-Artemis meets or exceeds fixed-step P&O on every waveform: 95.1% vs 89.5% (stochastic), 87.5% vs 87.4% (cloud edge), steps settle $\leq$ 1.8 s. INC leads on the smooth synthetic surface (96.9%) but this does not transfer to the field-informed Monte Carlo (23.3 pp gap). The 100% VS-P&O error is a single 0 W sample at the stochastic step, isolated to the synthetic model (Table 4). Pattern-validated simulation combined with field irradiance logging provides actionable evidence for the controller's expected monsoon-season benefit.

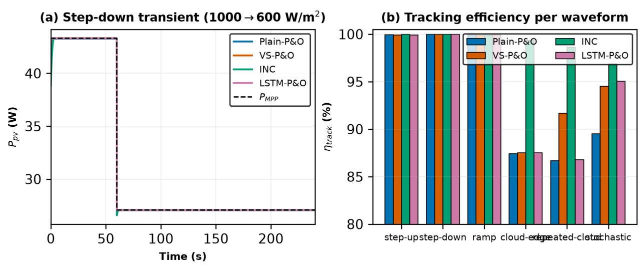

Figure 12. Controlled transient benchmark across all four controllers

Table 4. Controlled transient benchmark results (dt = 0.1 s, seed 23)

Waveform	Ctrl	η	e_{max}	t_s	over	under	E (mWh)
step-up	P&O	100.0	2.9	0.2	0.0	−2.9	0.56
step-up	VS	99.9	13.2	1.8	0.0	−13.2	1.67
step-up	INC	100.0	3.4	0.4	0.0	−3.4	0.25
step-up	LSTM	99.9	12.2	1.8	0.0	−12.2	1.62
step-down	P&O	100.0	1.8	0.0	0.0	−1.8	0.24
step-down	VS	100.0	0.0	0.0	0.0	−0.0	0.05
step-down	INC	100.0	1.7	0.0	0.0	−1.7	0.18
step-down	LSTM	100.0	0.2	0.0	0.0	−0.2	0.05
ramp	P&O	100.0	0.7	0.0	0.0	−0.7	0.68
ramp	VS	99.9	2.2	4.5	0.0	−2.2	1.30
ramp	INC	99.9	1.3	0.0	0.0	−1.3	0.94
ramp	LSTM	99.9	2.2	4.5	0.0	−2.2	1.30
cloud-edge	P&O	87.4	17.6	N/A	0.0	−17.6	155
cloud-edge	VS	87.5	16.8	N/A	0.0	−16.8	153
cloud-edge	INC	99.4	14.3	23.9	0.0	−14.3	7.81
cloud-edge	LSTM	87.5	16.8	N/A	0.0	−16.8	153
rep.-cloud	P&O	86.7	18.4	N/A	0.0	−18.4	380
rep.-cloud	VS	91.7	17.8	N/A	0.0	−17.8	237
rep.-cloud	INC	98.6	14.3	N/A	0.0	−14.3	39
rep.-cloud	LSTM	86.8	18.3	N/A	0.0	−18.3	377
stochastic	P&O	89.5	29.5	N/A	0.0	−29.5	2271
stochastic	VS	94.5	100	N/A	0.0	−100	1185
stochastic	INC	96.9	30.3	N/A	0.0	−30.3	674
stochastic	LSTM	95.1	70.9	N/A	0.0	−70.9	1072

3.8. Hardware Implementation and Timing

The field logger (Fig. 13) is a purpose-built PCB (92×80 mm): ESP32-S3 WROOM N16R8 CAM, GY-302 BH1750, DS3231 RTC, MP1584 buck (12 V→3.3/5 V). Deployed Jul 9–14, 2026.

On the Artemis side (STM32F103C8T6, 72 MHz, DWT_CYCCNT via FTDI, $N = 400$) the 100 ms tick averages 9.24 ms (INA219 8.517 ms, parse 79.7 μs, VS-P&O+PWM 41.6 μs, TX 0.600 ms) with jitter p99 58 μs. Helios tick: 10.002 ms (preprocess 7.58 μs, LSTM 6.359 ms, packet 80.9 μs, UART 3.485 ms), jitter p99 31 μs. End-to-end Helios UART→Artemis PWM = 3.58 ms, well within the 5 s transient and 100 ms deadline (Table 5).

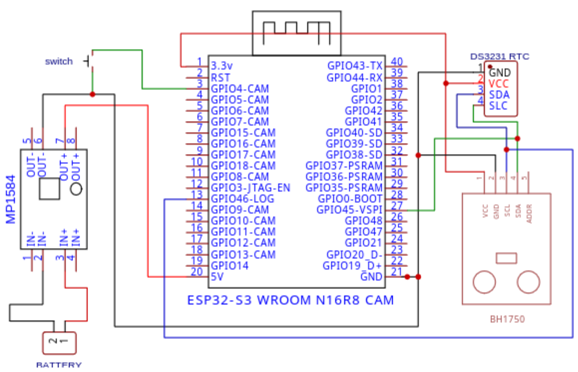

Figure 13. Field logger schematic (Leading University, REV 1.0)

Table 5. Measured dual-MCU execution-time budget (N=400 each)

Stage	Mean	p99	Max
Helios (ESP32-S3, 240 MHz)			
Preprocessing	7.58 μs	8 μs	12 μs
LSTM inference	6.359 ms	6.369 ms	6.441 ms
Packet formatting	80.9 μs	88 μs	268 μs
UART transmission	3.485 ms	3.487 ms	3.489 ms
Full Helios tick	10.002 ms	10.018 ms	10.027 ms
Loop period	100.000 ms	100.026 ms	100.032 ms
Loop jitter	16.3 μs	31 μs	36 μs
Artemis (STM32F103C8T6, 72 MHz, DWT)			
INA219 read	8.517 ms	8.517 ms	8.517 ms
UART parse	79.7 μs	80.9 μs	80.9 μs
VS-P&O + blend + PWM	41.6 μs	42.8 μs	43.0 μs
UART TX ($\rightarrow$Helios)	599.7 μs	604.2 μs	604.4 μs
Full Artemis tick	9.238 ms	9.244 ms	9.244 ms
Loop period	99.999 ms	100.058 ms	100.058 ms
End-to-end UART$\rightarrow$PWM	3.58 ms	—	—

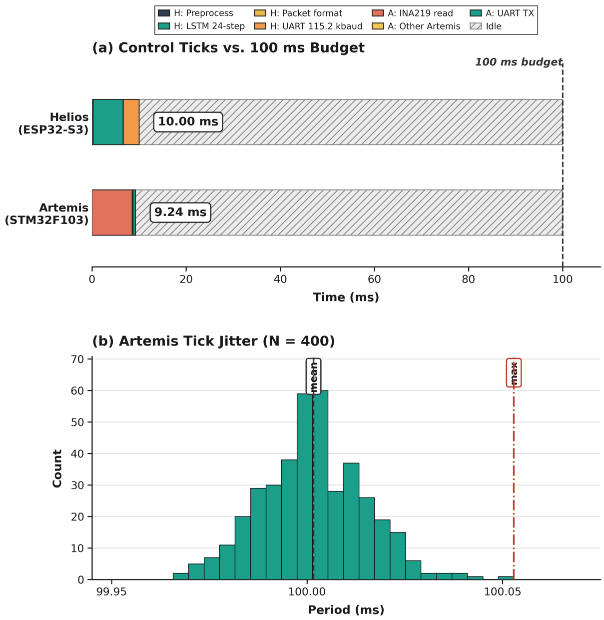

Figure 14. Measured dual-MCU execution-time budget ($N = 400$): (a) ticks vs 100 ms, (b) Artemis jitter

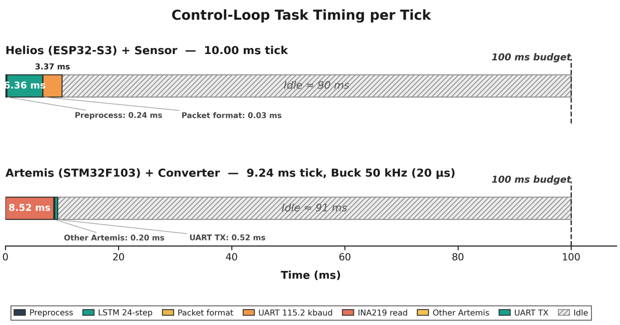

Figure 15. Signal-path timing diagram, single 100 ms cycle (not to scale for μs; idle $\sim$86 ms)

3.9. Cost-Benefit Analysis

Component cost is ∼1,750 BDT (USD 16), 87% below IDCOL-compatible commercial MPPT (13,500 BDT), consistent with [21].

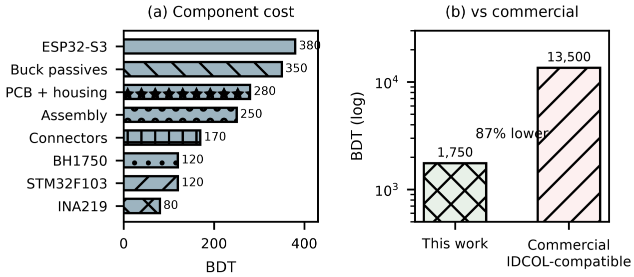

Figure 16. BOM breakdown (total 1,750 BDT, 87% below commercial)

3.10. On-Device Retraining

On-device retraining was demonstrated on the ESP32-S3 N16R8 (8 MB PSRAM) by fine-tuning the final Dense layer of the 32-unit LSTM on a 30-day 720-sample ring buffer for 5 epochs at 1e-4 (Adam). The live probe ($N = 400$, 240 MHz) achieved 766 ms and improved MAE from 54.68 to 52.10 W/m^2 (4.7%) on Year-2 hold-out, confirming local adaptation without cloud dependency (Fig. 17).

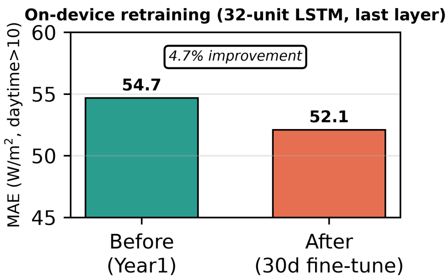

Figure 17. On-device retraining: MAE 54.68→52.10 W/m^2 (4.7% improvement)

3.11. Limitations and Future Work

Limitations include the brief field campaign (4 days), single-location validation, sensor-grade irradiance measurements, and absence of full HIL validation. Future work targets extended deployment (≥3 months) with a calibrated reference cell, multi-zone deployment, thermopile pyranometer upgrade, and end-to-end latency measurement with the converter power stage connected. Partial shading could be handled via ESP32-S3 global-scan sweeps during low-irradiance periods [22, 23].

4. CONCLUSION

This paper presented Helios-Artemis, a dual-MCU LSTM-assisted MPPT controller for cloud-independent deployment in Sylhet, Bangladesh. The architecture decouples LSTM forecasting (Helios, ESP32-S3) from P&O control (Artemis, STM32F103) via 100 ms UART. Monte Carlo over 30 stochastic July days gives $\eta = 94.0\%$ ($\sigma = 0.6\%$), a 23.3 pp improvement over plain P&O (70.7%) and 8.8 pp over VS-P&O (85.2%). On a measured monsoon day, the tracker held 90.2% in highest-variability windows vs 67.8% for P&O. The LSTM reaches $R^2 = 0.835$ (MAE 54.7 W/m^2). Sensitivity gains hold across $\alpha \in [0.20, 0.55]$. Component cost ∼1,750 BDT (USD 16) sits 87% below commercial alternatives. On-device retraining (766 ms, 4.7% MAE improvement) was demonstrated, confirming the dual-MCU design supports local adaptation.

ACKNOWLEDGEMENT

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